

RBE 502 – Homework 2

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All code is in the `matlab/` folder of the repo (one file per problem, `hw2_p1.m` to `hw2_p6.m`). The snippets below are only the key parts.

Problem 1

For each system I linearized at the origin, $A = \partial f / \partial x|_{x=0}$, and looked at the eigenvalues. If every $\text{Re}(\lambda) < 0$ the origin is asymptotically stable. The $x_1 x_2$ and x_2^3 terms drop out of A since their slope is zero at 0, and in systems 2 and 3 the factors $(1 - x_1^2 - x_2^2)$ and $(1 - x_1^2)$ just evaluate to 1 after the product rule.

	A	char. polynomial	eigenvalues
1	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	$(\lambda + 1)^2$	$-1, -1$
2	$\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$	$\lambda^2 + 2\lambda + 2$	$-1 \pm j$
3	$\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$	$\lambda^2 + \lambda + 1$	$-\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$
4	$\begin{bmatrix} -1 & -1 \\ 2 & 0 \end{bmatrix}$	$\lambda^2 + \lambda + 2$	$-\frac{1}{2} \pm j\frac{\sqrt{7}}{2}$

All real parts are negative, so the origin is asymptotically stable for all four systems.

```
J = jacobian(f, x);
J_at_origin = subs(J, x, [0; 0]);
A{i} = double(J_at_origin);
eigenvalues = eig(A{i});
```

Problem 2

Set $f(x) = 0$ to get the equilibria, then linearize at each one and check the eigenvalues of the Jacobian.

System 1. $\dot{x}_2 = 0$ gives $x_2 = 3x_1$, so $x_1(4 - x_1^2) = 0$ and $x_1 \in \{0, \pm 2\}$. With $J = \begin{bmatrix} 1 - 3x_1^2 & 1 \\ 3 & -1 \end{bmatrix}$:

- $(0, 0)$: $\lambda = \pm 2$, saddle.
- $(\pm 2, \pm 6)$: $\lambda^2 + 12\lambda + 8 = 0$, $\lambda = -6 \pm 2\sqrt{7}$, stable node.

System 2. From $x_2(x_1 - x_2) = 0$: either $x_2 = 0$ (then $x_1 = 2$ or -4) or $x_2 = x_1$ (then $x_1 = 2$ or 4). $J = \begin{bmatrix} 2x_1 - 2x_2 + 2 & -2x_1 + 4 \\ x_2 & x_1 - 2x_2 \end{bmatrix}$.

equilibrium	eigenvalues	type
$(2, 0)$	$6, 2$	unstable node
$(-4, 0)$	$-6, -4$	stable node
$(2, 2)$	$2, -2$	saddle
$(4, 4)$	$-1 \pm j\sqrt{7}$	stable focus

System 3. $x_2(3 - x_1 - 2x_2) = 0$ and $x_1(2 - x_1 - x_2) = 0$ give $(0, 0)$, $(2, 0)$, $(0, \frac{3}{2})$ and $(1, 1)$.

$$J = \begin{bmatrix} -x_2 & 3 - x_1 - 4x_2 \\ 2 - 2x_1 - x_2 & -x_1 \end{bmatrix}.$$

equilibrium	eigenvalues	type
$(0, 0)$	$\pm\sqrt{6}$	saddle
$(2, 0)$	$-1 \pm j$	stable focus
$(0, \frac{3}{2})$	$-\frac{3}{4} \pm j\frac{\sqrt{15}}{4}$	stable focus
$(1, 1)$	$-1 \pm \sqrt{2}$	saddle

System 4. Let $a(x) = \frac{1}{2} \tan(\pi x/2)$. The equilibrium conditions are $x_2 = a(x_1)$ and $x_1 = a(x_2)$. Subtracting them gives $x_2 + a(x_2) = x_1 + a(x_1)$, and since $x + a(x)$ is strictly increasing on $(-1, 1)$ this forces $x_1 = x_2 = x$ with $\tan(\pi x/2) = 2x$. The roots are $x = 0, \pm \frac{1}{2}$ (it is convex for $x > 0$ with a negative slope at 0, so there is only one positive root). The Jacobian is

$$J = \begin{bmatrix} -c & 1 \\ 1 & -c \end{bmatrix}, \quad c = \frac{\pi}{4} \sec^2\left(\frac{\pi x}{2}\right),$$

with eigenvalues $-c \pm 1$.

- $(0, 0)$: $c = \pi/4 < 1$, so $\lambda = -\frac{\pi}{4} \pm 1 = 0.215, -1.785$, saddle.
- $(\pm \frac{1}{2}, \pm \frac{1}{2})$: $c = \pi/2$, so $\lambda = -0.571, -2.571$, stable node.

I only looked at $|x_i| < 1$, the principal branch of $\tan$. The field has more equilibria on the other branches (for example $x_1 = x_2 \approx 2.89$), and they are all stable nodes since $c > 1$ there.

```
sol = solve(f == 0, [x1 x2]); % systems 1-3
xe = double([sol.x1, sol.x2]); % one equilibrium per row
A = double(subs(J, x, xe(k,:).'));

g = @(t) t - 0.5*tan(pi*t/2); % system 4, with x1 = x2 = t
r = fzero(g, [xs(k) xs(k+1)]); % bracket from a sign change
```

Problem 3

Same approach as Problem 1, just with three states.

1. $A = \text{diag}(-1, -1, 1)$. The $-x_1^2$ in $\dot{x}_3$ has zero slope at 0. One eigenvalue is $+1$, so the origin is **unstable**.
2. $A = \begin{bmatrix} -1 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$. It is block lower triangular, so $\det(\lambda I - A) = (\lambda + 1)(\lambda^2 + \lambda + 1)$ and $\lambda = -1, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$. **Asymptotically stable**.
3. $A = \text{diag}(-2, -1, -1)$. **Asymptotically stable**.

```
A = double(subs(jacobian(f, x), x, [0; 0; 0]));
re = real(eig(A));
if all(re < -tol), disp('asymptotically stable');
elseif any(re > tol), disp('unstable');
else, disp('inconclusive'); end
```

Problem 4

For $u = -Kx$ I wrote $A_{cl} = A - BK$ and matched $\det(sI - A_{cl})$ against the desired polynomial.

(a) Linear system

$\det(\lambda I - A) = (\lambda - 1)^2 - 9$, so $\lambda = 4, -2$ and the open loop is unstable. The controllability matrix is $[B \ AB] = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$ with determinant 3, so the system is controllable.

With $A_{cl} = \begin{bmatrix} 1 - k_1 & 3 - k_2 \\ 3 & 1 \end{bmatrix}$,

$$\det(sI - A_{cl}) = s^2 + (k_1 - 2)s - k_1 + 3k_2 - 8.$$

The desired poles $-1 \pm 2j$ give $s^2 + 2s + 5$. So $k_1 - 2 = 2$ gives $k_1 = 4$, and $-4 + 3k_2 - 8 = 5$ gives $k_2 = 17/3$.

$$K = \begin{bmatrix} 4 & \frac{17}{3} \end{bmatrix}$$

(b) DC motor

With the given numbers $A = \begin{bmatrix} -4 & 0.2 \\ 10 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ (I used A as printed in the assignment, with $+K/L$ in the top right). Then $\lambda^2 + 6\lambda + 6 = 0$, so $\lambda = -3 \pm \sqrt{3}$ and the open loop is asymptotically stable. $[B \ AB] = \begin{bmatrix} 2 & -8 \\ 0 & 20 \end{bmatrix}$ has determinant 40, so it is controllable.

$$A_{cl} = \begin{bmatrix} -4 - 2k_1 & 0.2 - 2k_2 \\ 10 & -2 \end{bmatrix} \text{ gives}$$

$$\det(sI - A_{cl}) = s^2 + (6 + 2k_1)s + 6 + 4k_1 + 20k_2.$$

The desired poles $-5 \pm 5j$ give $s^2 + 10s + 50$. So $k_1 = 2$, and $6 + 8 + 20k_2 = 50$ gives $k_2 = 9/5$.

$$K = \begin{bmatrix} 2 & 1.8 \end{bmatrix}$$

If the motor is meant to have $-K/L$ in that entry instead, the same steps give $k_2 = 1.6$.

```
Ac1 = A - B*[k1 k2];
char_cl = expand(det(s*eye(2) - Ac1));
char_des = pd(1)*s^2 + pd(2)*s + pd(3);
sol = solve(coeffs(char_cl - char_des, s), [k1 k2]);
K = place(A, B, p); % check
```

`place` returns the same gains, and the closed-loop eigenvalues come out at the requested poles.

Problem 5

State is $z = [x, \theta, \dot{x}, \dot{\theta}]^T$, and I use $D = I(M + m) + Mml^2 = (m + M)(I + ml^2) - m^2l^2$.

1. Small-angle linearization

With $\cos \theta \approx 1$, $\sin \theta \approx \theta$ and $\dot{\theta}^2 \sin \theta \approx 0$ the equations become

$$(m + M)\ddot{x} + ml\ddot{\theta} = F, \quad (I + ml^2)\ddot{\theta} + ml\ddot{x} = mgl\theta.$$

Solving the two for the accelerations:

$$\ddot{x} = \frac{(I + ml^2)F - m^2gl^2\theta}{D}, \quad \ddot{\theta} = \frac{mgl(M + m)\theta - mlF}{D}.$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{m^2gl^2}{D} & 0 & 0 \\ 0 & \frac{mgl(M+m)}{D} & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \frac{I+ml^2}{D} \\ -\frac{ml}{D} \end{bmatrix}$$

2. Nonlinear form and Taylor expansion

Solving the exact equations for $\ddot{x}$ and $\ddot{\theta}$, with $\Delta(\theta) = D + m^2l^2 \sin^2 \theta$:

$$\ddot{x} = \frac{(I + ml^2)F + ml(I + ml^2)\dot{\theta}^2 \sin \theta - m^2gl^2 \sin \theta \cos \theta}{\Delta(\theta)},$$

$$\ddot{\theta} = \frac{ml[-F \cos \theta + (M + m)g \sin \theta - ml\dot{\theta}^2 \sin \theta \cos \theta]}{\Delta(\theta)},$$

and $\dot{z} = [\dot{x}, \dot{\theta}, \ddot{x}, \ddot{\theta}]^T$. At $z = 0$, $F = 0$ both numerators are zero, so the derivative of Δ does not matter, and $\partial\Delta/\partial\theta = 0$ there anyway. The terms with $\dot{\theta}^2$ vanish as well. What is left is

$$\frac{\partial \ddot{x}}{\partial \theta} = -\frac{m^2gl^2}{D}, \quad \frac{\partial \ddot{\theta}}{\partial \theta} = \frac{mgl(M + m)}{D}, \quad \frac{\partial \ddot{x}}{\partial F} = \frac{I + ml^2}{D}, \quad \frac{\partial \ddot{\theta}}{\partial F} = -\frac{ml}{D},$$

which is the same A and B as in part 1. The code also checks this symbolically.

3. State feedback

With $m = 0.16$, $M = 0.48$, $l = 0.5$, $g = 9.8$, $I = \frac{1}{12}ml^2$:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -2.94 & 0 & 0 \\ 0 & 23.52 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 2.03125 \\ -3.75 \end{bmatrix}.$$

The open-loop eigenvalues are $0, 0, \pm 4.85$, so it is unstable. The controllability matrix has full rank (4), so poles can be placed anywhere. I picked $\{-2, -3, -4, -5\}$, i.e. $s^4 + 14s^3 + 71s^2 + 154s + 120$, and got

$$K = [-3.265 \quad -26.974 \quad -4.190 \quad -6.003], \quad F = -Kz.$$

I also simulated the full nonlinear model with this controller from $\theta_0 = 0.1$ and 0.3rad , and both go back to zero (Fig. 1).

Symbolic gain (bonus). Write $a = -\frac{m^2gl^2}{D}$, $b = \frac{mgl(M+m)}{D}$, $c = \frac{I+ml^2}{D}$, $d = -\frac{ml}{D}$. Expanding $\det(sI - A + BK)$ gives

$$s^4 + (ck_3 + dk_4)s^3 + (-b + ck_1 + dk_2)s^2 + (ad - bc)k_3s + (ad - bc)k_1,$$

and $ad - bc = -glm/D$. Matching to $s^4 + p_3s^3 + p_2s^2 + p_1s + p_0$:

$$k_1 = -\frac{Dp_0}{glm}, \quad k_3 = -\frac{Dp_1}{glm}, \quad k_2 = \frac{p_2 + b - ck_1}{d}, \quad k_4 = \frac{p_3 - ck_3}{d}.$$

Plugging in the numbers gives the same K as `place`.

```
A0 = simplify(subs(jacobian(f, z), vars, [0 0 0 0 0])); % Taylor
B0 = simplify(subs(diff(f, F), vars, [0 0 0 0 0]));
K = place(A0, B0, [-2 -3 -4 -5]);
cp = expand(det(s*eye(4) - (A0 - B0*K)));
Ks = solve(coeffs(cp - cdes, s), [k1 k2 k3 k4]); % symbolic gains
```

4. (Bonus) Equations of motion

Take the pendulum center of mass at $(x + l \sin \theta, l \cos \theta)$. Then

$$T = \frac{1}{2}(M + m)\dot{x}^2 + ml\dot{x}\dot{\theta} \cos \theta + \frac{1}{2}(I + ml^2)\dot{\theta}^2, \quad V = mgl \cos \theta,$$

and the generalized forces are $Q_x = F$, $Q_\theta = 0$. Lagrange's equation for x gives $(M + m)\ddot{x} + ml\ddot{\theta} \cos \theta - ml\dot{\theta}^2 \sin \theta = F$. For θ the $ml\dot{x}\dot{\theta} \sin \theta$ terms cancel and you get $(I + ml^2)\ddot{\theta} + ml\ddot{x} \cos \theta - mgl \sin \theta = 0$. Both match the given equations.

5. (Bonus) Linearization at $\theta = \pi$

Let $\varphi = \theta - \pi$, so $\sin \theta \approx -\varphi$ and $\cos \theta \approx -1$:

$$(m + M)\ddot{x} - ml\ddot{\varphi} = F, \quad (I + ml^2)\ddot{\varphi} - ml\ddot{x} + mgl \varphi = 0.$$

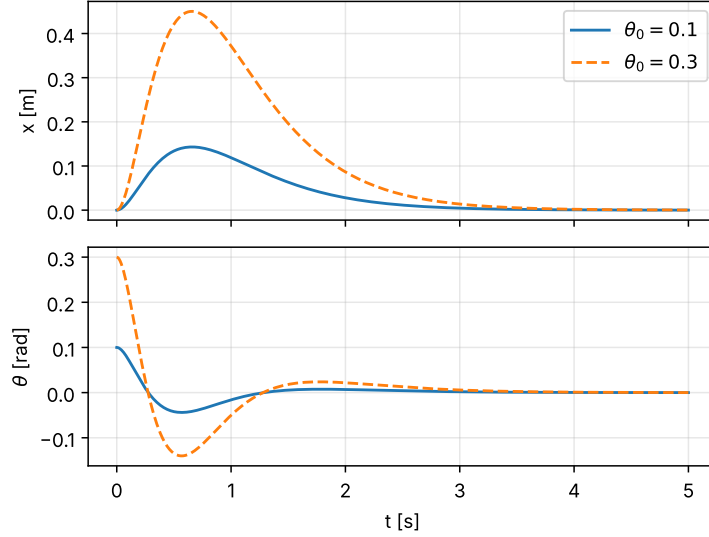


Figure 1: Nonlinear cart-pendulum with $F = -Kz$.

Solving gives

$$\ddot{x} = \frac{(I + ml^2)F - m^2gl^2\varphi}{D}, \quad \ddot{\varphi} = \frac{mlF - mgl(M + m)\varphi}{D}.$$

So A is the same as before except the $(4, 2)$ entry flips sign, and the last entry of B is $+ml/D$. The Taylor expansion gives the same thing. Now the lower block is oscillatory, $\lambda = \pm j4.85$, which makes sense for a pendulum hanging down.

Problem 6

The model is $\dot{x} = \nu \cos \theta$, $\dot{y} = \nu \sin \theta$, $\dot{\theta} = \omega$, $\dot{\nu} = u_1$, $\dot{\omega} = u_2$, with $V = \mathbf{x}^T \mathbf{x}$ where $\mathbf{x} = [x, y, \theta, \nu, \omega]^T$.

1. Derivative of V

$$\dot{V} = 2\mathbf{x}^T \dot{\mathbf{x}} = 2[x\nu \cos \theta + y\nu \sin \theta + \theta\omega + \nu u_1 + \omega u_2].$$

2. The given control law

Putting in $u_1 = -k_1\nu - x \cos \theta - y \sin \theta$ and $u_2 = -k_2\omega - \theta$:

$$\nu u_1 = -k_1\nu^2 - x\nu \cos \theta - y\nu \sin \theta, \quad \omega u_2 = -k_2\omega^2 - \theta\omega,$$

so the cross terms cancel and

$$\dot{V} = -2k_1\nu^2 - 2k_2\omega^2 \leq 0.$$

V is positive definite and radially unbounded, so the origin is stable and trajectories stay bounded. But $\dot{V}$ is only negative semidefinite (it is zero whenever $\nu = \omega = 0$), so this does not give asymptotic stability. I used LaSalle.

On $\{\dot{V} = 0\} = \{\nu = \omega = 0\}$, staying there requires $\dot{\omega} = u_2 = -\theta = 0$, so $\theta \equiv 0$, and $\dot{\nu} = u_1 = -x \cos \theta - y \sin \theta = -x = 0$, so $x \equiv 0$. Nothing forces y to zero, so the largest invariant set is

$$\mathcal{M} = \{(0, y, 0, 0, 0) : y \in \mathbb{R}\}.$$

Every point in $\mathcal{M}$ is an equilibrium of the closed loop, so trajectories converge to *some* point on that line but not necessarily the origin.

Conclusion: this controller does not asymptotically stabilize the origin.

Quantitative. The equilibrium set is a whole line, so the origin is not isolated and cannot be asymptotically stable. Linearizing the closed loop at 0 gives $\ddot{x} = -k_1\dot{x} - x$, $\ddot{\theta} = -k_2\dot{\theta} - \theta$ and $\dot{y} \approx 0$, so the eigenvalues are the roots of $s^2 + k_1s + 1$, $s^2 + k_2s + 1$ and an extra $s = 0$ from y . In simulation with $k_1 = k_2 = 1$ (Fig. 2) the states x, θ, ν, ω all go to zero but y ends at 0.664, -0.949 and 1.435 for the three initial conditions I tried, and V decreases but does not reach zero.

Qualitative. The robot cannot move sideways, so y only changes when $\nu \sin \theta \neq 0$. The controller drives θ and ν to zero, which freezes y wherever it happens to be. This is the usual nonholonomic problem: no smooth static feedback can asymptotically stabilize this kind of system (Brockett), so you need something time-varying or discontinuous.

```
Vdot = simplify(expand(2*q.'*f_cl)); % -2*k1*nu^2 - 2*k2*om^2
[t, S] = ode45(dyn, [0 200], inits{i}, opts);
```

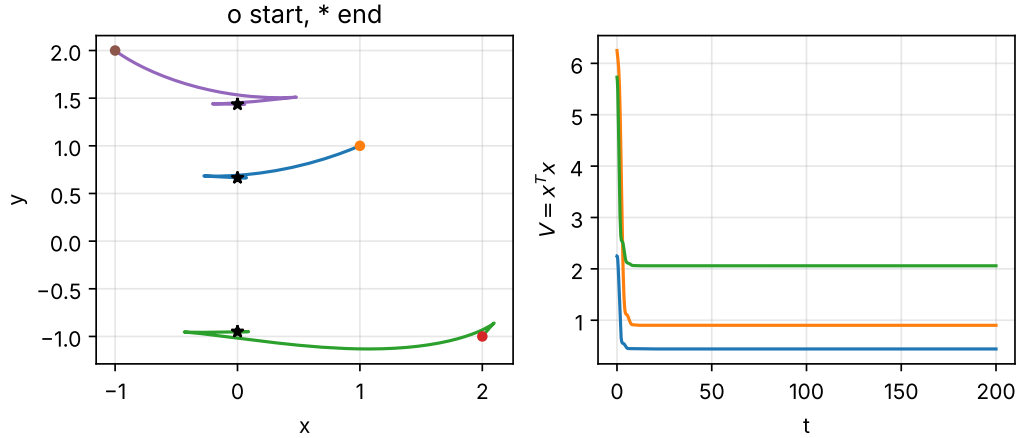


Figure 2: Closed-loop trajectories end on the line $x = \theta = 0$ at $y \neq 0$ (left); V does not go to zero (right).